

9.2.7

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

Find the area of the region bounded by the parabola $y^2 = x$ and the line $2y = x$.

Solution:

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (1)$$

$$\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix}, \quad f = 0 \quad (2)$$

$$(3)$$

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}, \quad \kappa \in \mathbb{R} \quad (4)$$

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad (5)$$

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \quad (6)$$

Substituting eq-(6) in eq-(1) and Solving the Quadratic

$$\kappa_i = \frac{1}{\mathbf{m}^T \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{(\mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u}))^2 - g(\mathbf{h})(\mathbf{m}^T \mathbf{V} \mathbf{m})} \right) \quad (7)$$

$$g(\mathbf{h}) = \mathbf{h}^T \mathbf{V} \mathbf{h} + 2\mathbf{u}^T \mathbf{h} + f \quad (8)$$

$$\mathbf{V} \mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{m}^T \mathbf{V} \mathbf{m} = 1 \quad (9)$$

$$\mathbf{V} \mathbf{h} + \mathbf{u} = \begin{pmatrix} -\frac{1}{2} \\ 0 \end{pmatrix}, \quad \mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u}) = -1 \quad (10)$$

$$g(\mathbf{h}) = 0 \quad (11)$$

$$\kappa_i = -(-1) \pm \sqrt{(-1)^2 - 0} = 1 \pm 1 \quad (12)$$

$$\implies \kappa_1 = 0, \quad \kappa_2 = 2 \quad (13)$$

$$\mathbf{x}_1 = \mathbf{h} + 0\mathbf{m} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \mathbf{h} + 2\mathbf{m} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \quad (14)$$

$$\text{Area} = \int_0^2 (2y - y^2) dy = \left(y^2 - \frac{y^3}{3} \right)_0^2 = 4 - \frac{8}{3} = \frac{4}{3} \quad (15)$$

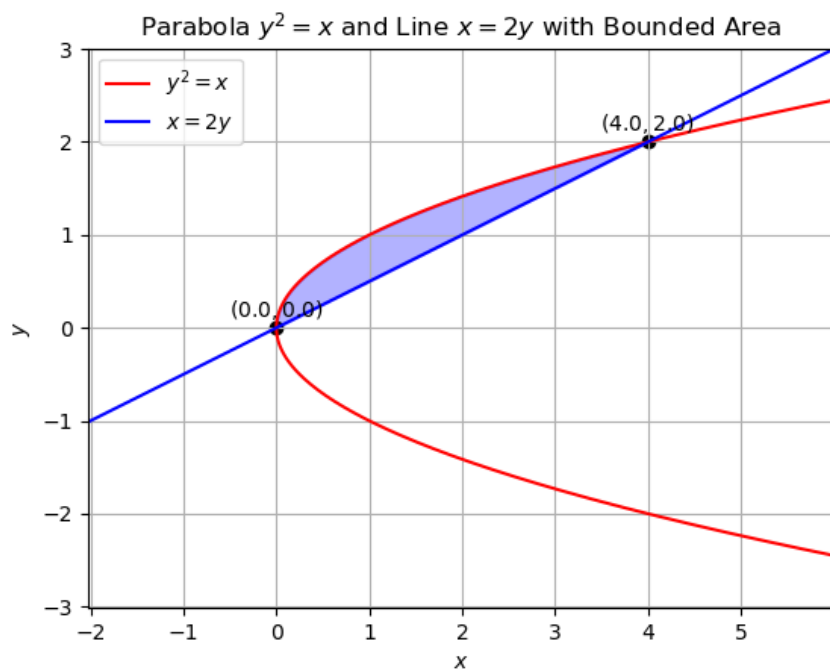


Figure 1